

LYNCHBURG, VA, USA

WMO: 724100

Lat: 37.321N

Lon: 79.207W

Elev: 940

StdP: 14.20

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13733

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.7	18.8	-3.3	4.8	22.0	1.3	6.1	25.1	19.4	47.4	17.6	45.5	3.7	300	0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	91.7	73.5	89.5	72.9	87.2	72.0	76.4	86.7	75.2	84.9	74.2	83.3	5.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	128.4	80.1	72.3	123.9	79.0	71.4	119.9	78.0	40.3	87.1	39.2	85.2	38.2	83.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	14.9	12.5	5.2	95.0	6.1	3.0	0.8	97.2	-2.7	99.0	-6.1	100.7	-10.6	102.9
			4.7	79.0	5.1	1.5	1.0	80.1	-1.9	81.0	-4.8	81.8	-8.5	82.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.5	35.9	39.1	46.8	56.9	64.5	72.3	75.8	74.7	68.2	57.3	46.7	39.4
	DBStd	16.03	10.16	9.36	9.56	8.34	7.24	5.32	4.42	4.42	6.20	7.93	8.57	8.78
	HDD50	1515	451	323	179	28	1	0	0	0	0	23	163	347
	HDD65	4216	901	726	567	267	102	9	1	1	39	261	549	793
	CDD50	3905	15	16	79	234	450	669	800	765	545	249	64	19
	CDD65	1133	0	0	2	23	86	228	336	301	134	22	1	0
	CDH74	9863	1	2	40	315	801	1984	2952	2519	1035	201	12	1
	CDH80	3502	0	0	4	68	208	686	1209	955	329	43	0	0
(14) Wind	WSAvg	4.8	5.5	5.6	6.2	6.3	5.1	4.3	3.8	3.6	4.1	4.2	4.6	4.7
(15) Precipitation	PrecAvg	40.8	3.0	3.1	3.7	3.1	3.9	3.3	4.2	3.4	3.4	3.1	3.0	3.2
	PrecMax	59.7	8.0	6.1	9.3	8.0	9.1	10.0	10.3	11.3	12.0	11.4	8.8	7.2
	PrecMin	26.5	0.5	0.6	0.7	0.3	0.9	0.5	1.1	0.9	0.0	0.0	0.3	0.3
	PrecStd	8.7	1.7	1.5	1.8	1.5	2.0	2.3	2.1	2.0	2.7	2.6	1.8	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.6	70.7	78.7	86.1	88.4	92.8	95.7	94.1	90.9	84.8	75.5	69.5
		MCWB	58.2	57.6	60.4	64.4	70.5	73.9	74.8	74.6	70.8	69.4	61.4	57.3
	2%	DB	62.0	64.4	73.4	81.7	85.1	89.5	92.2	91.0	87.3	79.4	69.8	63.3
		MCWB	56.0	52.9	57.1	62.4	69.1	72.8	74.1	73.5	70.7	65.8	58.0	56.7
	5%	DB	57.5	59.8	68.4	77.7	82.2	86.8	89.5	88.3	83.9	75.0	66.0	58.5
		MCWB	51.4	50.2	54.5	60.8	67.7	72.1	73.6	72.9	69.3	63.6	56.9	52.1
	10%	DB	52.3	54.8	63.6	73.0	78.8	84.1	86.8	85.5	80.4	71.2	61.9	54.3
		MCWB	46.1	46.3	52.6	58.6	66.3	70.8	72.8	71.8	68.0	62.5	54.4	48.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.8	61.2	62.9	68.2	73.3	76.3	78.1	78.1	75.3	72.6	65.1	63.7
		MCDB	65.2	66.9	74.5	78.2	83.3	87.1	89.5	88.8	84.2	78.1	68.4	67.6
	2%	WB	57.5	56.5	60.2	65.5	71.4	74.9	76.6	76.2	73.4	69.8	62.2	58.4
		MCDB	61.4	61.4	69.2	75.8	81.4	85.5	87.2	86.2	81.4	75.4	67.1	61.4
	5%	WB	52.1	51.9	57.6	63.3	69.7	73.6	75.4	74.9	71.8	67.0	59.1	53.6
		MCDB	55.8	57.3	65.3	73.2	79.0	83.7	85.2	84.1	79.1	72.9	64.1	56.9
	10%	WB	46.3	47.0	54.3	60.9	68.0	72.6	74.3	73.6	70.3	63.7	55.4	48.5
		MCDB	51.6	53.4	61.9	69.8	76.1	81.7	83.5	81.8	76.6	68.9	60.4	52.9
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	19.4	21.5	23.2	21.5	20.5	19.8	19.8	20.0	21.6	21.5	18.2
		MCDBR	24.2	27.6	28.3	29.3	25.5	23.7	23.5	23.6	24.9	25.9	27.0	24.0
		MCWBR	17.8	18.6	16.0	13.7	11.8	10.0	8.3	8.9	10.3	13.3	16.5	17.1
	5% WB	MCDBR	21.6	23.9	24.3	24.2	21.8	20.9	20.5	20.1	19.9	20.1	21.1	20.8
		MCWBR	18.0	18.3	15.4	13.2	10.7	9.4	8.2	8.6	9.2	11.9	16.0	17.1
(40) Clear-Sky Solar Irradiance	taub		0.299	0.315	0.348	0.388	0.430	0.476	0.506	0.506	0.399	0.357	0.318	0.302
	taud		2.510	2.454	2.408	2.314	2.211	2.125	2.064	2.040	2.337	2.443	2.515	2.537
	Ebn at Noon		284	292	291	283	272	258	249	246	270	272	274	275
	Edh at Noon		25	30	35	40	45	49	52	52	37	30	25	23
(44) All-Sky Solar Radiation	RadAvg		746	981	1311	1626	1806	1968	1856	1680	1392	1119	834	648
	RadStd		56	117	85	124	157	134	117	83	138	131	63	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.61	N/A	N/A	N/A	N/A	N/A	N/A	+227	+148
(47) Regional (46 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	+252	+140

Nomenclature: See separate page